

Academic Hour 1

Introduction to WordPress Plugins

What is a WordPress Plugin?

A **plugin** is a small piece of software that **adds functionality to WordPress without modifying the core system**.

Examples of plugins:

- Contact forms
 - SEO tools
 - Security plugins
 - Page builders
 - E-commerce (WooCommerce)
-

Why Plugins Are Important

Plugins allow developers to:

- Extend WordPress
- Add new features
- Customize websites
- Avoid modifying WordPress core files

Important rule:

! Never modify WordPress core files

Always use:

- Plugins
- Themes
- Hooks

How Plugins Work

WordPress plugins work using **hooks**.

Two main types:

Actions

Execute code at a specific moment.

Example:

```
add_action()
```

Filters

Modify existing data.

Example:

```
add_filter()
```

For this beginner lesson we will focus on **actions**.

Creating Your First Plugin

Step 1 — Open WordPress plugin directory

Navigate to:

```
wp-content/plugins
```

Step 2 — Create a new folder

Example:

```
academy-first-plugin
```

Step 3 — Create the plugin file

Inside the folder create:

academy-first-plugin.php

Plugin Header (Required)

Every WordPress plugin must start with a **plugin header comment**.

Example:

```
<?php
/*
Plugin Name: Academy First Plugin
Description: My first custom WordPress plugin.
Version: 1.0
Author: Student
*/
```

After saving this file:

Go to

WordPress Admin → Plugins

You will see your plugin in the list.

Activate it.

Example Plugin 1

Display Message in Footer

Now we will add a message at the bottom of the website.

Add this code below the header:

```
function academy_footer_message() {
```

```
    echo "<p style='text-align:center;'>This website was built with WordPress!</p>";  
}  
  
add_action('wp_footer', 'academy_footer_message');
```

What this code does

Function:

academy_footer_message()

prints HTML text.

The hook:

wp_footer

tells WordPress to run the function **in the website footer**.

Result

When students refresh the website they will see:

This website was built with WordPress!

displayed in the footer.

Practical Task (10 minutes)

Students should:

- 1 Create their own plugin folder
- 2 Create the plugin PHP file
- 3 Add plugin header
- 4 Activate the plugin
- 5 Add their own custom message

Example message:

Hello from my first WordPress plugin!

Academic Hour 2

Adding More Plugin Features

Now students understand the basics.

Let's extend our plugin.

Example Plugin 2

Add a Dashboard Notice

We will display a message inside the **WordPress admin dashboard**.

Add this code to the plugin:

```
function academy_admin_notice() {  
    echo "<div class='notice notice-success'>  
        <p>Welcome! Your custom plugin is active.</p>  
    </div>";  
}  
  
add_action('admin_notices', 'academy_admin_notice');
```

What Happens Now?

When admin opens the dashboard they will see:

Welcome! Your custom plugin is active.

inside the admin panel.

Example Plugin 3

Create a Simple Shortcode

Shortcodes allow users to insert content inside pages.

Example:

[academy_message]

Add this code:

```
function academy_shortcode_message() {  
    return "<h3>This message was added using a shortcode!</h3>";  
}
```

```
add_shortcode('academy_message', 'academy_shortcode_message');
```

How to Use the Shortcode

Open a page or post.

Write:

[academy_message]

The message will appear on the page.

Practical Task (15 minutes)

Students must:

Create a plugin that:

- ✓ Shows a footer message
- ✓ Shows a dashboard notice
- ✓ Creates a shortcode

Shortcode name:

[student_message]

Example output:

Hello! I created my first WordPress plugin.

Mini Project (20 minutes)

Students must create a plugin called:

academy-tools

Plugin must include:

Feature 1

Footer message

Feature 2

Admin dashboard notice

Feature 3

Custom shortcode

Example Final Plugin Structure

academy-tools/
academy-tools.php

Teacher Tip

Ask students:

- What happens if we deactivate the plugin?
- Why are plugins better than editing themes?
- What other features could plugins add?

This helps them understand **WordPress architecture**.

Quick Quiz (End of Lesson)

1

What is a WordPress plugin?

- A. A theme
 - B. A feature extension for WordPress
 - C. A database
 - D. A server
-

2

Where are plugins located?

- A. wp-admin
 - B. wp-content/plugins
 - C. wp-includes
 - D. themes
-

3

What is a WordPress hook?

- A. A CSS class
 - B. A way to run code at a specific time
 - C. A database table
 - D. A theme option
-

4

Which hook runs code in the footer?

- A. wp_footer
 - B. wp_head
 - C. init
 - D. admin_init
-

5

What does a shortcode do?

- A. Adds CSS
 - B. Displays dynamic content inside pages
 - C. Deletes posts
 - D. Changes themes
-

Learning Outcome

Students now know how to:

- ✓ Create plugins
- ✓ Use hooks
- ✓ Extend WordPress functionality
- ✓ Build reusable code